

- Q1. (4 marks)** Let $\mathcal{U} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set. Consider the following sets:

$$F = \{4, 5, 9, 10\}$$

$$G = \{1, 2, 4, 6, 9\}$$

$$H = \{1, 4, 5, 7, 9\}$$

List all elements in each of the following sets. **For each part, list the elements in at least one other set that you used along the way.** For example, in (a), you could list the elements of $\overline{G}$.

(a) $\overline{G} \cap H$

(b) $G \cap (F \cup H)$

(c) $\overline{(G \cup H)}$

(d) $(F \cap H) - G$

- Q2. (4 marks)** Prove the following statement.

$$\forall x \in \mathbb{N}, \exists y \in \mathbb{N}, \quad (x^2 + 2 - y) \mid (10 - x^2 - y^2)$$

- Q3. (5 marks)** Prove the following statement. **Cite each theorem as you use it.**

$$S : \text{For all integers } p, q, r, \text{ if } p^2q \mid (r^2 - q) \text{ and } pq \nmid (q(7 + 5r) - 7r^2) \text{ then } pq^3 \nmid qr.$$

Citing a theorem could look like “By Pythagorean Theorem,”

- Q4. (7 marks)** Consider the universal set $\mathcal{U} = \mathbb{Z}$, and the sets $\mathcal{F} = \{f \in \mathbb{Z} : 42 \mid (f + 18)\}$, $\mathcal{K} = \{3k : k \in \mathbb{Z}\}$ and $\mathcal{P} = \{p \in \mathbb{Z} : 7 \mid (p + 18)\}$. For each of the following statements, clearly state whether it is true or false. Then prove or disprove it.

(a) $\forall x \in \mathbb{Z}, \left[(x \in \mathcal{F}) \Rightarrow (x \in (\mathcal{K} \cap \mathcal{P})) \right]$

(b) $\forall x \in \mathbb{Z}, \left[(x \notin \mathcal{F}) \Rightarrow (x \notin (\mathcal{K} \cap \mathcal{P})) \right]$

Q5. (5 marks) Hint: For this question, you may cite the following without proving it.

Proposition PS: if m^3 is a perfect square, then m is a perfect square.

(a) Prove that the following sets are not disjoint.

$$K = \{(2s)^2 : s \in \mathbb{Z} \text{ and } 3000 \leq s \leq 4000\} \text{ and } L = \{2u^3 : u \in \mathbb{Z}\}$$

(b) **Indicate whether the following statement is true or false, and then prove or disprove it.**

$$|K \cap L| = 1.$$

Note: Your method should not rely on verifying more than 30 cases/numbers.

Q6. (5 marks) For each of (a) - (d), indicate a domain, S , and an open sentence, $P(x, y)$, from the following tables to make a true statement. In your solution, you must use each domain and each open sentence **exactly once**. You do not need to provide proofs that these statements are true.

Domain, S	Open sentence, $P(x, y)$
$\mathcal{J} = \{j \in \mathbb{Z} : j \geq 0\}$	$P_1(x, y) : x^2 \geq y$
$\mathcal{E} = \{2k : k \in \mathbb{Z}\}$	$P_2(x, y) : \frac{xy}{2} \in S$
$\mathcal{D} = \{d \in \mathbb{Z} : 2 \nmid d\}$	$P_3(x, y) : (x^2 + 3) \mid (y^2 + 4)$
$\mathcal{M} = \{m^2 : m \in \mathbb{Z}\}$	$P_4(x, y) : xy > 0$

(a) $\forall x \in S, \forall y \in S, P(x, y)$ with domain $\underline{\quad S \quad}$ and open sentence $\underline{\quad P(x, y) \quad}$.

(b) $\forall x \in S, \exists y \in S, P(x, y)$ with domain $\underline{\quad S \quad}$ and open sentence $\underline{\quad P(x, y) \quad}$.

(c) $\exists y \in S, \forall x \in S, P(x, y)$ with domain $\underline{\quad S \quad}$ and open sentence $\underline{\quad P(x, y) \quad}$.

(d) $\exists x \in S, \exists y \in S, P(x, y)$ with domain $\underline{\quad S \quad}$ and open sentence $\underline{\quad P(x, y) \quad}$.